

Neuro-Symbolic Search for Binary Linear Systems: LP-Guided MCTS with Null-Space Projection and Policy-Guided Complete Backtracking

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Laser NDT image recovery is a binary linear system that demands both feasibility detection and exact solutions

Task: find $x \in \{0,1\}^n$ with $Ax = b$, or certify that none exists

Noisy scans (dust) make the system infeasible — a false negative discards a recoverable scan

Market-split-type systems: NP-complete; LP-based branch-and-bound degrades sharply with m

Our approach: stage cheap null-space certificates before learned search, and turn search failures into proofs

Kernel pump: alternating projection onto $\{x : Ax = b\}$ — walks the null space, certifies feasibility constructively

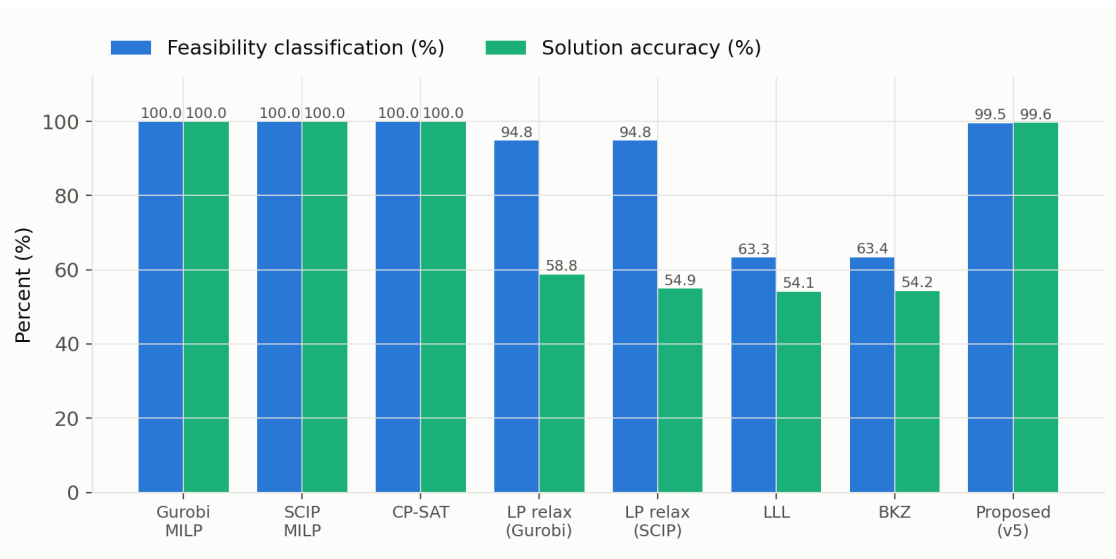
AlphaZero-aligned MCTS: policy trained on full visit distributions + root Dirichlet noise

Policy-guided DFS: complete backtracking under the learned ordering — solution or infeasibility proof

Inference spends symbolic certificates first and reserves learned search for the hard residue

- 1 Constraint propagation + probing** → contradiction = infeasibility proof
- 2 LP hard rule** → LP infeasible $\Rightarrow$ binary infeasible
- 3 Kernel pump** → exact solution = feasibility certificate
- 4 GNN feasibility head** (threshold calibrated on training data only)
- 5 MCTS + 1-ply backtracking + pump repair**
- 6 Policy-guided DFS** (20k-node budget) → solve or prove infeasible

The pipeline beats every inexact baseline: 99.5% feasibility, 99.6% solution accuracy



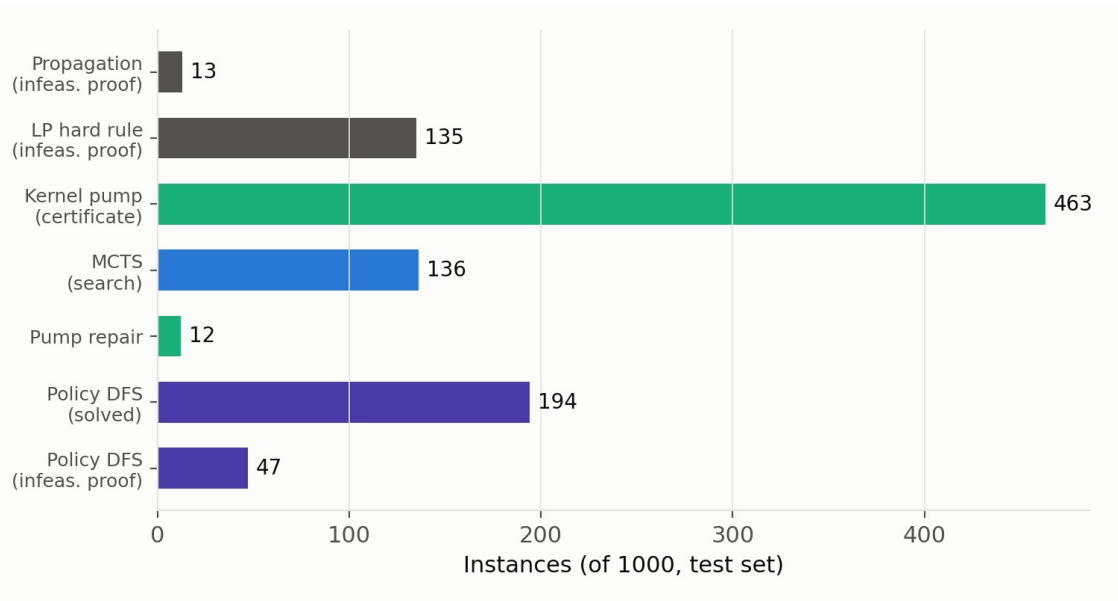
Key finding

Zero false negatives
(800/800)

+40.9 pp solution accuracy
vs. best LP rounding

Exact solvers stay perfect
and $\sim 10^3\times$ faster at 10×25
— our case rests on scaling,
not this size

61% of instances close before any neural inference; the learned policy closes most of the rest



Attribution

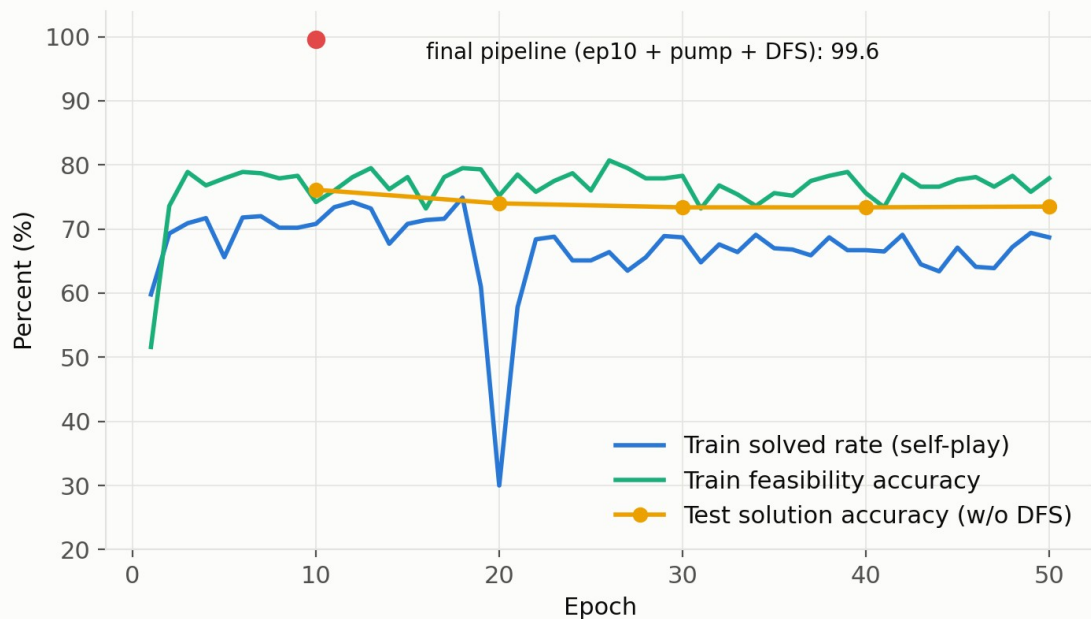
Symbolic proofs: 148 infeasible

Kernel pump: 463 solved (0.78 ms each)

Learned search: 128 MCTS + 194 DFS solved, 47 DFS infeasibility proofs

Residue: 5 FP + 3 unsolved (budget)

Test accuracy peaks at epoch 10 and drifts down — a fixed training pool induces memorization



Interpretation

Train metrics recover after the ep-20 collapse, test does not follow

Checkpoint chosen on a training-derived calibration set

Next cycle: held-out eval + on-the-fly instance generation

Proofs beat probabilities: the learned feasibility head contributed zero correct rejections

Hard cases (LP-feasible, binary-infeasible) are inseparable to the head — probabilities cluster above 0.84 for both classes

Every feasibility gain came from proofs: propagation, pump certificates, DFS exhaustion

Learning helps inside the search (ordering, value) — consistent with the learned-branching literature

Null-space structure plus budgeted complete search met the operational targets; size scaling is the next milestone

Targets met: feasibility 99.50% (≥ 95), solution 99.62% (≥ 75) on held-out test

Honest bound: exact solvers remain perfect and faster at 10×25

Next: feasibility-head redesign, on-the-fly instances, size-generalization study

References

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